Faculty of Information Technology and Artificial Intelligence
Network Administration (10686213)
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Linux Command Practice in Ubuntu WSL

Lecture 3 Commands and a Local AI Assistant

1. Introduction

I practiced the Linux commands from Lecture 3 in Ubuntu running through Windows Subsystem for Linux (WSL2). The exercises covered shell navigation, manual pages, files and directories, text viewing and processing, and a small AI-assisted command task. I ran the commands in a dedicated practice folder named `network-admin-lab`.

2. Environment

Operating environment	Ubuntu 26.04.1 LTS on WSL2
Linux account	cys_oday
Practice directory	/home/cys_oday/network-admin-lab
System check	uname -a identified the Microsoft WSL2 kernel

3. Navigation and File Management

I used **`pwd`** to confirm the current location and **`cd`** to move into `files`, to the parent directory, to my home directory, and back to the lab folder. The path examples included `.`, `..`, and `~`. I then practiced listing, copying, moving, and removing sample files and directories.

Commands practiced: **`ls, cd, pwd, mkdir, cp, cp -r, mv, rm, rmdir`**.

The file copy placed `fruits.txt` in `files/`. A recursive copy of `source-dir` created `source-copy/` with `note.txt` inside. I moved the copied fruit file to `renamed-fruits.txt`, removed it, and removed the empty `empty-dir` directory. The listings confirmed each change.

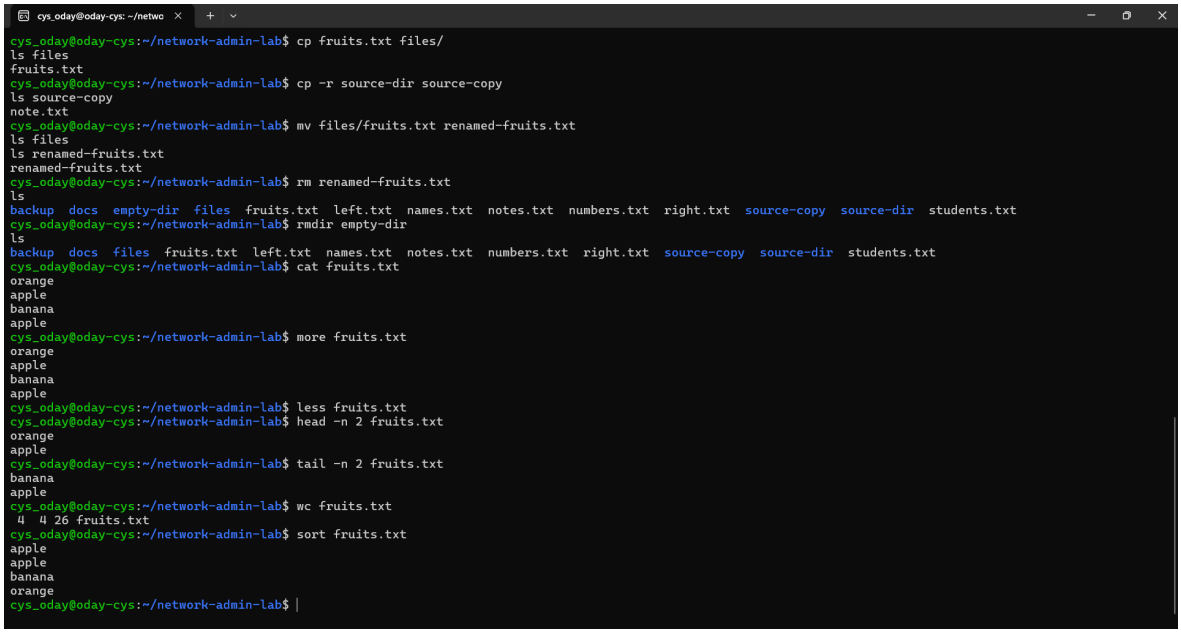


Figure 1. File and directory operations, followed by sample file viewing and sorting.

4. Viewing and Processing Text

The `fruits.txt` sample contained orange, apple, banana, and apple on separate lines. I viewed it with **`cat`**, **`more`**, and **`less`**; **`head -n 2`** showed the first two lines, and **`tail -n 2`** showed the last two. **`wc`** reported 4 lines, 4 words, and 26 bytes.

`sort fruits.txt` ordered the entries alphabetically, **`sort -r fruits.txt`** reversed that order, and **`sort fruits.txt | uniq`** removed adjacent duplicates after sorting. For colon-separated sample records, **`cut -f 1 -d:`** selected the names and **`sort -t: -n -k 3`** sorted by the third field numerically. **`paste left.txt right.txt`** joined matching lines; **`grep apple`** selected the two matching fruit lines; **`tr`** converted the fruit names to uppercase.

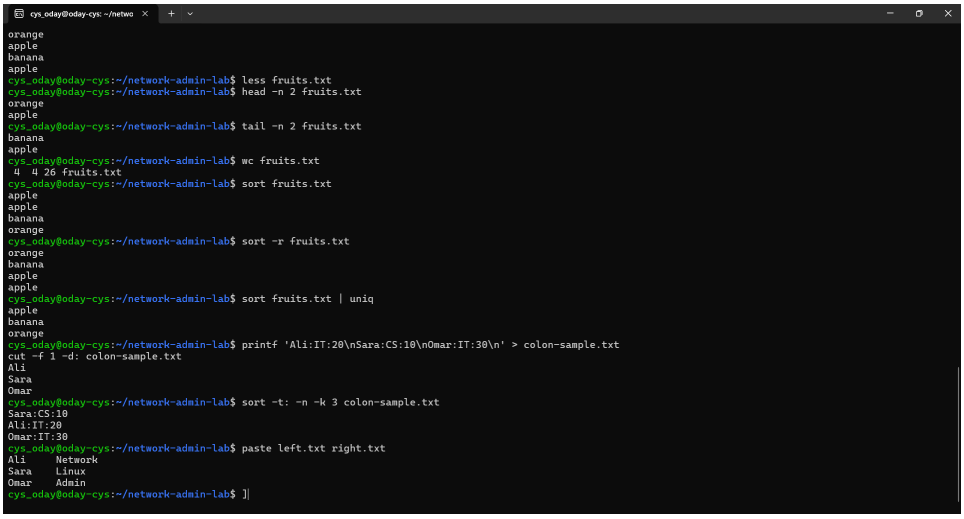


Figure 2. Text processing results, including sort, uniq, cut, numeric field sorting, and paste.

5. Manual Pages and vi

I opened **`man ls`** and **`man 1 intro`**, and searched manual descriptions with **`man -k copy`**. I also tried **`man -K "copy files"`**; it stepped through many matching pages, so I stopped the search with Ctrl+C. In **`vi`**, I created `lab-note.txt` and saved it with **`ZZ`**. I accidentally typed **`:wq`** while still entering text, so those characters became part of the note. This helped me distinguish vi's command mode from text-entry mode.

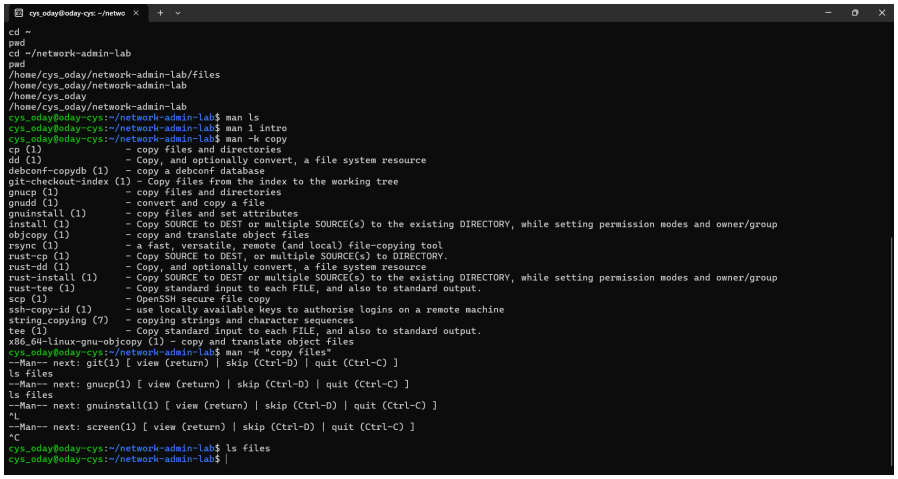


Figure 3. Directory navigation and manual-page searches in the lab terminal.

6. Local AI Assistant Trial

I set up Ollama and downloaded Qwen2.5:3b, then installed ShellGPT with LiteLLM in a Python virtual environment and configured it to use the local Ollama service. A short prompt test returned a response from the model.

Prompt: `Create a file named ai-demo.txt in the current directory containing the line Created by local AI in Ubuntu WSL.`

ShellGPT proposed: **`echo "Created by local AI in Ubuntu WSL" > ai-demo.txt`**. I reviewed the command and chose the execute option. Then **`cat ai-demo.txt`** displayed the expected line, confirming that the generated command ran successfully.

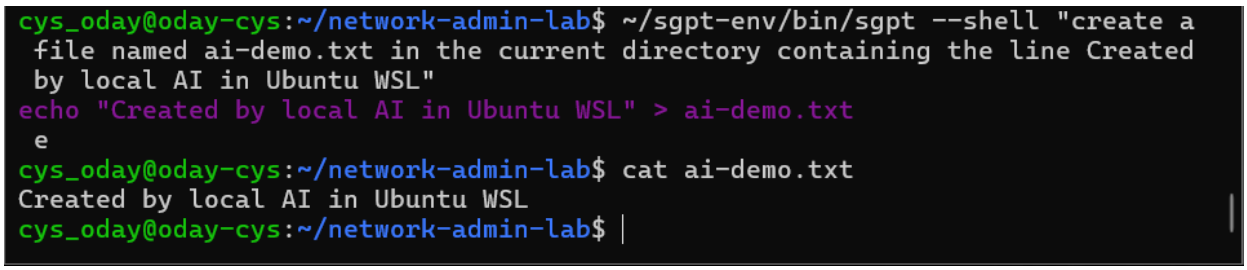


Figure 4. ShellGPT's suggested command, approval, execution, and verified file contents.

7. Conclusion

The practice improved my familiarity with common Linux file and text commands and with using manual pages to look up command documentation. The local AI assistant produced a correct command for a simple file-creation request. Reviewing the suggested command before approving execution helped me understand what it would change.

References

Lecture 3 - Intro to Linux, course slides.
Ollama Linux installation: https://ollama.com/download/linux
Qwen2.5 model library: https://ollama.com/library/qwen2.5
ShellGPT Ollama setup: https://github.com/TheR1D/shell_gpt/wiki/Ollama